Vivado 2024.2: On-Chip Power Dissipation & Energy (reports/pnr_power_100mhz.rpt)

| Tool Version : Vivado v.2024.2 (lin64) Build 5239630
| Design : pulp_riscv_ai_soc_top | Target: xc7z010clg400-1 (Speed -1)
| Command : report_power -file synth/reports/pnr_power_100mhz.rpt

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1. On-Chip Power Summary (@ 100.0 MHz, 25.0°C Typical)

Total On-Chip Power (W) : 0.158 | Total Heat Sink Power : 0.158 W
Dynamic Power (W) : 0.065 | <-- Mini-PULP Core Logic + HWPE: 65.0 mW
Device Static Power (W) : 0.093 | <-- Zynq-7000 Baseline Quiescent: 93.0 mW
Effective TJA (C/W) : 11.5 | Junction Temperature : 26.8 °C
Max Ambient Allowed (C) : 83.2 | Thermal Safety Margin: 58.2 °C [SAFE]

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2. Energy Efficiency Assessment (Inference at 100 MHz):
*** Latency per Image : 127.96 us (12,796 clock cycles)**
* Active Energy/Image : 8.32 uJ (65.0 mW x 127.96 us) [Ultra-Low Energy]